

MOHAMMAD SAAD

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PROFESSIONAL SUMMARY

Senior Consultant with 4+ years of experience in technical consulting and hands-on data engineering. Currently at Protiviti, I assessed data architecture and DevOps practices for a major bank in the GCC, designing solutions for CI/CD pipelines, observability platforms, and real-time data integration. Previously at PwC, I built scalable AWS data pipelines that migrated multi-terabyte datasets and reduced processing time while maintaining data consistency. I worked with AWS services (Glue, S3, EMR, Lambda, Step Functions), Apache Spark, Python, and SQL. I am comfortable working with cross-functional teams, translating technical requirements into practical solutions, and delivering results for enterprise clients in financial services.

TECHNICAL SKILLS

- **Big Data & Analytics:** Apache Spark, PySpark, ETL Data Pipelines, Data Architecture, Distributed Computing, Real-time Data Processing, Batch Processing, Statistical Analysis, AVRO / PARQUET file formats Data Analysis
- **Cloud Platforms & Data Storage:** AWS (Glue, S3, Athena, EMR, RDS, Redshift, Lambda, Step Functions, CloudWatch, EventBridge, IAM), Data Warehousing, Data Lakes, RDBMS, NoSQL Databases, Change Data Capture CDC, Qlik, MongoDB
- **Programming & Scripting:** Python, SQL, PostgreSQL, Pandas, NumPy, AWS Wrangler, Data Transformation, Semi-structured Data Processing

EXPERIENCE

Senior Data & AI Consultant

Protiviti

September 2024 - Present

- Assessed DevOps maturity across 30 applications using capability models covering CI/CD pipelines, testing automation, security integration, and monitoring practices for a leading bank in KSA.
- Analyzed CI/CD pipeline implementations across 7 lifecycle phases (Plan, Development, Integration, Testing, Deployment, Operations, Monitoring).
- Developed observability correlation solutions for metrics, events, logs, and traces using Dynatrace, and custom instrumentation - reduced manual log correlation time from hours to minutes through automated trace ID propagation.
- Built standardized CI/CD processes supporting hybrid VM and container deployments with GitLab, Argo CD, quality gates for SAST/DAST scanning, and automated artifact management through Nexus repository.
- Designed SRE framework with SLA/SLO/SLI definitions, error budget management, and monitoring blueprints.
- Defined 200+ technical and business KPIs measuring deployment frequency, MTTR, test coverage, API response times, transaction success rates, and system availability across payment and banking applications.
- Architected target state technical solutions for automated deployment pipelines, centralized observability platforms, error budget tracking systems, and infrastructure-as-code implementations with specific tooling and integration requirements.
- Analyzed Microsoft Dynamics CRM with custom workflows, plugins, and validation scripts, identifying performance bottlenecks and batch processing limitations affecting daily transactions.
- Designed API-first integration architecture to replace SFTP/XML batch processing, defining services and event-driven microservices architecture for real-time data exchange with financial institutions.
- Evaluated technology and database solutions to address current system scalability constraints.
- Designed hybrid cloud deployment strategy maintaining local data residency compliance with disaster recovery capabilities.
- Assessed data architecture challenges with XML schema containing 100+ fields, proposing optimized data models and Change Data Capture (CDC) implementation for real-time synchronization.
- Contributed to RFP technical requirements defining system architecture, API specifications, security controls, and performance benchmarks for vendor evaluation process.

Data Engineer

PwC

October 2021 - September 2024

- Built comprehensive data quality monitoring and validation processes maintaining alignment between cloud databases and on-premises source systems. Achieved 99.9% data consistency across structured and semi-structured data sources.
- Created automated validation checks and anomaly detection to catch data quality issues early in the pipeline and prevent downstream errors.
- Generated daily statistical summaries and data quality reports comparing cloud and on-premises data. Provided actionable insights for data quality improvements.
- Migrated large-scale datasets (multi-terabyte volumes) to AWS S3 data lakes and RDS PostgreSQL databases using AWS Glue ETL jobs and EMR clusters with Apache Spark and PySpark.
- Built and optimized scalable ETL pipelines processing high-volume structured and semi-structured data from multiple source systems.
- Automated ETL pipeline operations through custom Python scripts and AWS Step Functions orchestration. Reduced manual intervention by 60% and data processing time by 40%.
- Performed comprehensive data analysis including source system identification, data profiling, data dictionary collection, and source-to-target mapping to meet business needs.
- Ran exploratory data analysis using Python (Pandas, NumPy) and SQL to understand data patterns, identify anomalies, and support data-driven decisions.
- Delivered daily operational reporting and statistical analysis using AWS Athena and custom PySpark scripts. Provided actionable insights for performance optimization.
- Monitored data pipeline performance using AWS CloudWatch and custom metrics. Troubleshoot data processing issues to maintain 99.5% uptime and high system reliability.
- Set up end-to-end data pipeline observability with automated alerting for data quality issues, processing failures, and performance problems.
- Used comprehensive AWS data services including Glue for ETL, S3 for storage, Athena for analytics, EMR for big data processing, Step Functions for orchestration, CloudWatch for monitoring, and EventBridge for event-driven processing.
- Worked with business stakeholders and development teams to understand data requirements and translate functional requirements into technical acceptance tests.
- Supported product owners and analysts in defining data quality standards and building fit-for-purpose validation frameworks.
- Used libraries like AWS Wrangler, Pandas, and NumPy to convert semi-structured inputs into structured outputs for analysis.

EDUCATION

Bachelor of Technology in Information Technology
Motilal Nehru National Institute of Technology Allahabad

CERTIFICATIONS

- AI Essentials – University of Oxford
- Agentic AI Developer Certification Program - Ready Tensor (Ongoing)
- Oracle Certified Foundations Associate - Oracle University,
- AWS Partner: AWS Cloud Practitioner Essentials - Skillbuilder,
- Microsoft Azure Essentials - Great Learning,
- Managing Machine Learning Projects with Google Cloud - Coursera,
- Introduction to Spark SQL and Data Frames - LinkedIn